commands). The techniques have been known since the 2017 paper and the release of open-source models such as LLaMa from Meta. The internet revolution enabled the availability of a large volume of data that was used to build both the proprietary models and open-source models. For example, the latest model behind ChatGPT used 45 Terabytes of data from the open internet. This class of models requires a type of computing system known as accelerated computing devices, such as Graphics Processing Units (GPUs) or Application Specific Integrated Circuits (ASICs) or more colloquially "accelerator chips", which enable trillions of calculations to be completed in parallel. Each accelerator chip can perform trillions of calculations per second, and many hundreds to thousands of these chips are needed to build GenAI models. To crunch through 45 TB of data, the ChatGPT model is rumoured to have run for over two months across 10,000-plus accelerator chips! To put this in perspective, every set of 1,000 accelerator chips requires roughly one megawatt (MW) of power. So, to build the models behind ChatGPT, over 10 MW of power was needed for over two months—the equivalent of powering 2,000-3,000 homes!

The power of GenAI models has grown in direct proportion to the accelerator chips available to build them—the more accelerator chips we throw at a model, the more powerful it tends to become. The US and China each have several hundred thousand high-powered accelerator chips working as AI supercomputers to build the foundation of their AI-driven economy and AI strategies. This insatiable need to consume accelerator chips has resulted in an arms race for stockpiling them. The problem is so acute that the US recently banned the sale of these chips to China, in the hopes of slowing down China's march towards AI domination.

The current crop of GenAI models are inherently biased—not intentionally, but due to the availability of data and compute. For example, when asked to generate an image of a successful business person, the models invariably generate an image with a white man in a business suit. The largest and most powerful models available today, such as GPT4 that powers ChatGPT and Microsoft Bing, Gemini that powers Google Bard, or LLaMa2 from Meta source less than one per cent of their data from India. Couple that with the statistic that less than one per cent of the world's accelerator chips are deployed in India, and it makes the inequity worrisome. To unlock the potential of AI in India and other countries, it is necessary to ensure fair representation.

Several governments, including the US and India as well as the EU bloc, have issued draft legislation aimed at the responsible use of AI, and specifically GenAI. Most regulations rely on self-reporting by GenAI software firms. There is a section of the community that has raised concerns about the potential for these models to exceed human capabilities and become "Artificially Generally Intelligent" (AGI), and pose a threat to humanity. This may at best give us an interesting academic debate. For, even though the technology may mimic some human tasks, it is far from being able to "think" like humans, let alone exceed their cognitive abilities. In my opinion, AGI is not a practical risk for the foreseeable future.

India has two urgent challenges to address—representation for Indian languages as well as data sourced from India in GenAI models and the severe shortage of accelerator chips. Efforts such as Bhashini driven by MeitY (the ministry of electronics and information technology) are a great start in building Indian language datasets. The private sector is also starting to drive GenAI development. Even if India addresses the data challenge, the dire lack of chips needs to be fixed to jump-start innovation. The accelerator chip arms race perhaps needs to be taken with the same seriousness as the nuclear arms race of decades past, where countries that take an active stance come out stronger, leaving nations with passive policies vulnerable. Unlike past technologies like nuclear energy that were predominantly advanced through government funding, GenAI development (outside China) has been led by private firms like Google, Microsoft and Meta. This asymmetric investment dynamic makes regulating it much harder. A public-private partnership in India can drive a more responsible GenAI development and adoption in India.

The world is still in the early stages of GenAI adoption and it is important for India to invest responsibly and at-scale in AI. I am optimistic that a public-private partnership will drive a Cambrian explosion of GenAI from India, providing new opportunities and uplifting the disenfranchised populations of India and the world. ■

TAKEAWAYS

- GenAI models have grown in proportion to the accelerator chips available, sparking a US-China arms race in chips
- The inequity is worrisome: less than one per cent of the world's accelerator chips are deployed in India
- The other urgent challenge for India is the severe shortage of data sourced from here in GenAI models